

Effects of Osmotic Stress on Adaptive Radiation and *wspF* Mutations in *Pseudomonas fluorescens*

Abstract

Adaptive radiation enables populations to diversify rapidly in response to ecological pressures. In microbial systems, this process is evident from observing colony phenotypes and analyzing associated genetic mutations. The study examined the effects of altered osmotic conditions at the air-liquid interface on adaptive radiation in *Pseudomonas fluorescens*. The hypothesis is that saline layering near the liquid-air interface would reduce phenotypic diversity and increase the number of required mutations necessary for the emergence of the wrinkly spreader (WS) phenotype.

P. fluorescens cultures were grown under three conditions: homogeneous, static, and experimental. Morphology for each growth type was quantified using the Shannon diversity index. WS isolates from the experimental and static colonies were sequenced for the *wspF* gene to identify mutations.

Static cultures showed the highest phenotypic diversity ($H' = 0.166$), followed by the experimental culture ($H' = 0.0805$), with the homogenous condition showing no diversification ($H' = 0$).

Sequencing indicated that the experimental isolate of WS contained more and a broader range of mutations in the *wspF* gene than the static isolate, including multiple frameshift mutations.

These results indicate that increasing the environmental stresses constrains both phenotypic diversification (H') as well as available pathways for adaptive radiation.

Introduction

Adaptive radiation is the process by which a single ancestral lineage rapidly diversifies into multiple distinct forms to inhabit different ecological niches (Britannica). An example of adaptive radiation is Darwin's finches. While voyaging on the HMS Beagle, Darwin observed that Galapagos finches evolved unique beak shapes depending on the island and environment in which they inhabited. Varying conditions and, most importantly, food sources imposed ecological stress, driving divergent selection and the evolution of a once-uniform population. Small mutations affecting finch beaks enabled these individuals to exploit specialized niches, giving them a minority advantage (Grant & Grant, 2008). Adaptive radiation can be defined by evolution driven by environmental opportunity, divergent selection, and mutations. Studying adaptive radiation provides a framework for understanding how ecological opportunity and natural selection shape phenotypic variation.

P. fluorescens is a Gram-negative bacterium most commonly found in plant rhizospheres, freshwater systems and many other high-oxygen environments (Spiers et al., 2003). It is

characterized by rapid growth, large population sizes, and the ability to adapt rapidly to ecological niches. *P. fluorescens* thus functions as an experimentally traceable model of adaptive radiation. Rainey & Travisano found that the SBW25 strain in the preticular region is particularly phenotypically plastic, rapidly generating ecological variants when exposed to new environments. This means that, despite evolution generally unfolding over thousands or millions of years, underlying forces such as environmental opportunity, divergent selection, and mutations allow us to observe evolution on a human timescale.

In static environments, oxygen diffuses downward from the water surface, creating a vertical oxygen gradient that provides different niches for *P. fluorescens* to exploit. Rainey and Travisano showed that, in the static environment described above, an initially uniform SBW25 population can diversify into three distinct morphs within less than 48 hours. Smooth (SM) ancestral type, wrinkly spreaders (WS), and fuzzy (fluffy) spreaders (FS) differ in how they occupy ecological niches in their environment. WS mutants form a biofilm at the liquid surface, allowing them to exploit oxygen-rich conditions. FS mutants exploit the temporary lapses in the WS biofilm by floating to the top, while the ancestral SM phenotype remains suspended in the liquid phase. However, when the cultures are repeatedly agitated and the oxygen gradient is eliminated, this diversification of phenotypes does not occur. This evidence indicates that these mutations constitute adaptive radiation.

The mutations that differentiate the phenotypes of *P. fluorescens* originate from the Wsp chemosensory pathway. This is encoded by the *wspA-F* operon and regulates intracellular levels of the secondary messenger cyclic-di-GMP, which controls biofilm formation (Hickman et al., 2005). A loss-of-function mutation in *wspF*, which acts as a negative regulator of the Wsp pathway, results in a constantly methylated *wspA* and, as a result, a constitutively active *wspR*. Constitutive activation of *wspR* results in elevated c-di-GMP levels, promoting excessive cellulose production and surface attachment at the air-liquid interface (Spiers et al., 2003).

The goal of this study was to determine how altering osmotic conditions at the liquid-air interface affects adaptive radiation in *P. fluorescens*. More specifically, we examined whether a layer of saline on the surface of our static test cultures would reduce phenotypic diversity and increase the mutations in *wspF* in the WS phenotype. We hypothesized that saline-treated static cultures would exhibit lower phenotypic diversity than untreated static controls, and that WS mutants arising under saline experimental conditions would have a greater number and breadth of *wspF* mutations, reflecting greater selective pressure on the Wsp pathway. Our work investigated how small environmental changes both constrain phenotypic diversification and the genomic pathways available for adaptive radiation.

Materials and Methods

Four sterile culture vials were prepared, each containing 3mL of tryptic soy broth (TSB). All vials were inoculated with a single colony of *P. fluorescens*. One vial was incubated under static conditions, one under constant agitation to create homogeneous conditions, and two under experimental conditions identical to the static treatment but with a 1mL saline solution layered on the surface. The four cultures were incubated at 28 °C for 72 hours.

Next, each culture was serially diluted to 10^{-7} and concentrations 10^{-5} , 10^{-6} , and 10^{-7} were plated onto tryptic soy agar (TSA). Plates from the 10^{-6} dilution were used to quantify colony morphologies.

WS colonies were isolated from both the static and experimental conditions and grown in TSB for 24 hours at 28 °C. Genomic information (DNA) was extracted using a column-based centrifuge extraction protocol. DNA concentration and purity were then measured using a NanoDrop spectrophotometer.

The *wspF* gene was amplified using PCR with OneTaq2x Master Mix (New England Biolabs). PCR products were run on a 3% agarose gel and visualized before submission for sequencing (Genewiz, Inc.). The resulting sequences were aligned to the ancestral *P. fluorescens* for mutation analysis.

Results

The experiment was divided into three different conditions. The homogeneous condition was maintained by continuously agitating the vial during incubation, thereby preventing the formation of an oxygen gradient. The static condition was a vial that was not moving, allowing an **oxygen concentration gradient** to form. Finally, the experimental condition was prepared as for the static condition, except that 1 mL of saline solution was floated on the surface of the vial.

For all three conditions, colony morphologies and the Shannon diversity index were recorded. Under both static and experimental conditions, *wspF* was PCR-amplified and sequenced.

All three expected phenotypes- Smooth spreaders (SM), Wrinkly spreaders (WS), and Fuzzy spreaders (FS)- were present in our experiment. The static condition yielded the most diversity with an H' of 0.166. In this condition, a 10^{-6} concentration of the static bacterial broth solution, the TSA plate yielded 2 WS and 1 FS. Static was also the only condition in which an FS was present. The next most phenotypically diverse condition was the experimental condition. The experimental condition had a Shannon index of 0.0805, a reduced but non-zero result. WS presence was reduced, as only 1 WS and no FS were present for this condition. Finally, the homogenous condition had an H' of 0, with all 52 colonies being of the ancestral SM phenotype.

The trend for the number of colonies from each condition was the same, with the static condition having the most at 91, followed by the experimental condition at 64, and the homogenous condition having the fewest colonies at 52.

Table 1: Phenotypes of *P. fluorescens* as a function of growth condition. The growth conditions (Static, Experimental, and Homogeneous) refer to the conditions under which the vials were incubated. The homogeneous condition was maintained by continuously agitating the vial during incubation; the static condition was maintained by keeping the vial on a stationary shelf; and the experimental condition was the same as static, but with 1 ml of saline layered on top. The morph frequencies were ascertained by counting TSA plates from a 10^{-6} dilution of the bacterial broth.

	Smooth	Wrinkly	Fuzzy
Homogenous	52	0	0
Static	88	2	1
Experimental #1	63	1	0

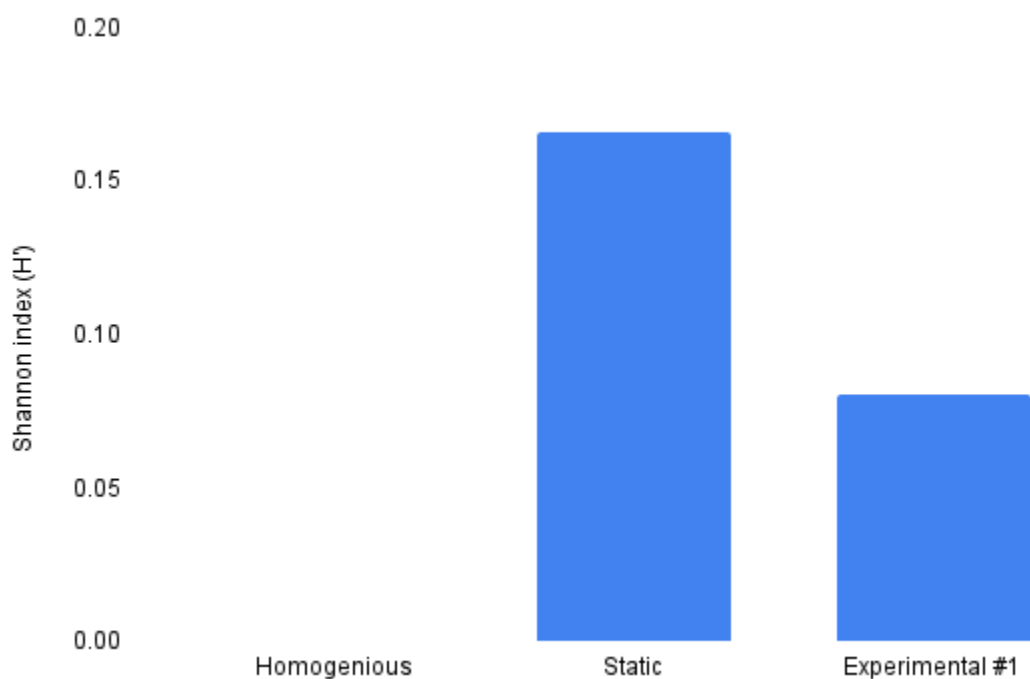


Figure 1: Shannon index as a function of growth condition. The growth conditions (Static, Experimental, and Homogeneous) refer to the conditions under which the vials were incubated. The homogeneous condition was maintained by continuously agitating the vial during incubation; the static condition was maintained by keeping the vial on a stationary shelf; and the experimental condition was the same as static, but with 1 ml of saline layered on top. The Shannon index was calculated from phenotype frequencies obtained by counting TSA plates at a 10^{-6} dilution of the bacterial broth.

PCR amplification of the *wspF* gene was successfully performed on WS isolates from static and experimental growth conditions. Agrose gel electrophoresis revealed a dominant strand at about

1050 base pairs, which is consistent with the ~1000 expected base pairs present from *wspF* amplification. Band sizes were consistent across isolates, but the intensity of the static isolate was significantly lower, indicating mostly reliable amplification of the *wspF* gene (fig. 2). The amplified *wspF* products were subsequently subjected to DNA sequencing to identify mutations associated with the WS phenotype.

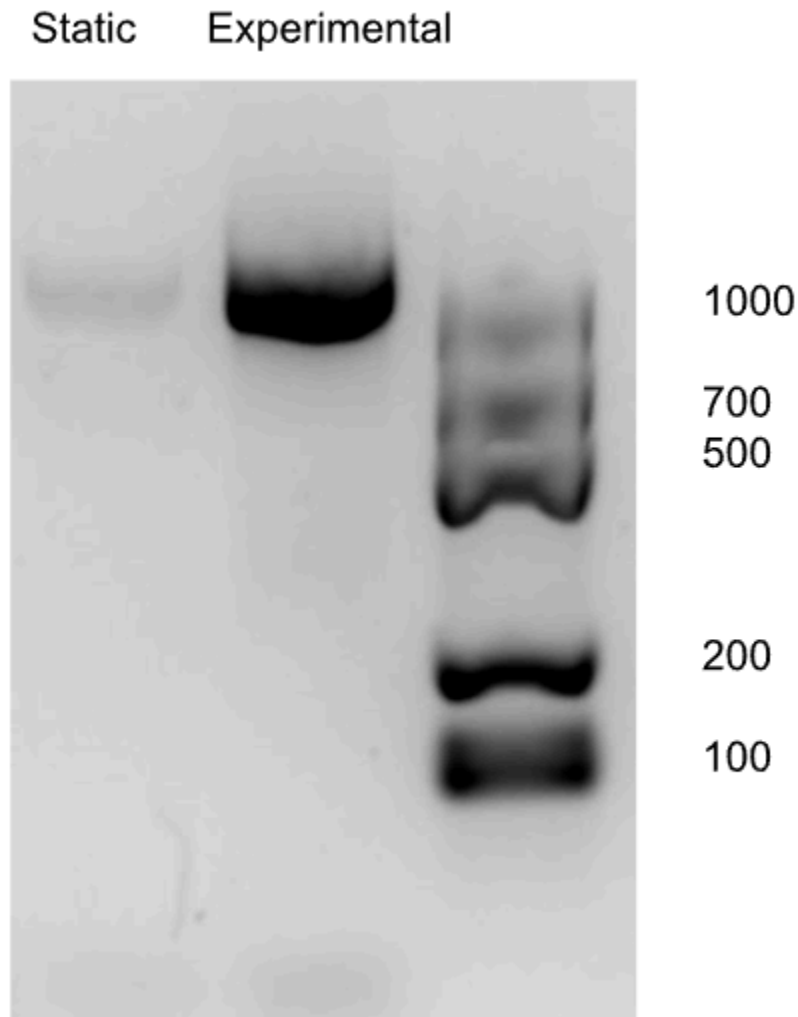


Figure 2: Agarose gel electrophoresis of *wspF* for static and experimental conditions.

The dominant band in both lanes corresponds to an expected ~1050 base pairs for the *wspF* gene. The more pronounced experimental band suggests a higher DNA yield. The ladder on the right provides a size reference in base pairs for comparison with the amplified fragments.

Sequencing of the *wspF* gene from WS isolates under experimental culture conditions revealed numerous nucleotide mutations relative to the ancestral SM sequence. Identified mutations included a silent C35A transversion, deletions spanning nucleotides 141-155 and 999-1011, a C243G transversion and a singular base insertion of G at position 265-266. Several of these mutations, most notably the deletion at positions 141-155, rendered the *wspF* coding region inoperable due to a frameshift (Table 2).

Sequencing of the *wspF* gene from WS isolates under static culture conditions revealed fewer nucleotide mutations relative to the saline-treated (experimental) isolates. Mutations identified in the static WS isolate included an insertion of 11 nucleotides in the position 125-126 as well as a G201A transition. The insertion at 125-126 caused a frameshift mutation in the *wspF* gene. The mutations observed in the WS static isolate were fewer in number and represented a narrower range of mutation types than those observed in the experimental isolate.

Table 2: *wspF* mutations from different condition WS genotypes.

The table presents the mutations identified in the sequenced *wspF* for the experimental and static WS. The table shows the nucleotide changes, the type of mutation, and the resulting effect on the amino acid chain. The Experimental condition showed a greater number and breadth of mutations than the static condition, including a deletion that caused a frameshift.

Condition	Nucleotide change	Nature of mutation	Amino acid change
Experimental	C35A	Transversion	Silent
	Δ141_155	Deletion	P47Δ(4)
	C243G	Transversion	
	265_266InsG	Insertion	
	Δ999_1011	Deletion	
Static	125_126InsCGCCGA ACTGA	Insertion	P47Δ(1)
	G201A	Transition	

Discussion

This study investigated how altering osmotic conditions at the liquid-air interface influences the adaptive radiation of *P. fluorescens*. We hypothesized that adding a saline layer to *P. fluorescens* cultures during incubation would reduce phenotypic diversity and alter the mutational landscape of the WS phenotype by increasing the number of mutations required. Both hypotheses were partially supported.

Static cultures exhibited the highest morphological diversity, consistent with previous work, indicating rapid niche partitioning in oxygen-gradient environments. In contrast, the saline-treated Experimental cultures showed significantly reduced diversity, with a complete lack of the FS morphotype (Figure 1). This suggests that increasing osmotic stress restricts the range of ecological niches to which *P. fluorescens* can adapt, thereby limiting observed adaptive radiation.

Sequencing results further support this interpretation. The experimental WS isolate contained a significant increase in mutations, including a greater variety of mutations in *wspF*, compared with the static isolate (Table 2). Since *wspF* functions as a negative regulator of the secondary messenger c-di-GMP, mutations in this regulator lead to constitutive activation of biofilm construction. The greater mutational burden observed in our experimental isolate suggests that the additional saline concentration imposes stronger selective constraints, requiring more genetic disruptions to activate the WS phenotype.

These results are consistent with the known molecular mechanism of the Wsp pathway. Elevated c-di-GMP levels promote cellulose production and surface attachment, permitting WS mutants to float near the surface and colonize the oxygen-rich air-liquid interface. However, the addition of saline likely reduced the fitness advantage of the oxygen-rich surface, creating less competitive environments for WS mutants and favouring only strongly biofilm-forming genotypes.

A limitation of the study is the low number of sequenced WS isolates, which restricts statistical power. In addition, because only *wspF* was sequenced, we ignored potential mutations in other regulatory genes, such as a gain-of-function mutation in *wspC*. Future studies could sequence the entire genome of Wsp or multiple WS isolates to increase statistical power, identify the most impactful mutations for the phenotype WS, and determine whether saline conditions bias the evolutionary trajectory toward a specific genetic pathway.

Overall, our study suggests that environmental structure can not only shape phenotypic diversity but also constrain the genetic pathways available for adaptive radiation. Even in simple microbial systems, such as *P. fluorescens*, small changes in ecological conditions can affect both evolutionary consequences and molecular mechanisms.

Acknowledgements

Thank you, Ms. Boylan, for your continued assistance on this journal, and Torri for her dedicated partnership.

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